PRE-IDEATHON TEMPLATE

GENESIS IDEATHON

TEAM & DOMAIN

- Team Name – HACKNOVA
- Team Members –

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- Selected Domain – #6
- Problem Statement – "When emergency strikes, the clock is the enemy. Families chase blood donors and funds through endless calls and messages. What if one platform could bring both to the bedside in minutes?"



DOMAIN PROBLEM IN OUR WORDS

What exactly needs to be solved?

- The main problem is the lack of coordination during medical emergencies.
  Families often have to search separately for blood donors, blood banks, financial help, and hospital information.
- Finding help takes too much time.
  Even if a compatible donor or financial support is available nearby, families may not know where to find them quickly.
- There is no single trusted place connecting all the resources. Blood donors, hospitals, blood banks, and people willing to contribute financially are often scattered across different platforms
- We need to turn a stressful, manual search into one simple emergency request that can quickly connect a patient with the right blood donors and verified financial support.

Why is it important right now?

- Medical emergencies are time-sensitive. Delays in finding blood or arranging treatment funds can make an already difficult situation even harder.
- People are already connected digitally, but that connectivity isn't always organized for emergencies.
- WhatsApp, calls and social media can spread requests quickly, but they don't provide proper verification, matching, or tracking.
- A technology-based solution can bring existing resources together and make help faster, more reliable, and easier to coordinate.

WHO IS AFFECTED?

Target users, stakeholders, beneficiaries

1. Primary Users / Target Group

Patients and Families Facing Medical Emergencies

- Individuals requiring urgent blood support due to road accidents, emergency surgeries, childbirth complications, or other critical medical conditions.
- Patients with chronic conditions, such as thalassemia, who require regular transfusions.
- Families seeking reliable information about blood availability and donor support.

Voluntary Blood Donors

- Individuals willing to donate blood but lacking a reliable, centralised platform to identify genuine and urgent requirements.
- Registered donors seeking timely notifications based on their blood group, availability, and proximity to participating hospitals or blood banks.

2. Secondary Stakeholders

Hospitals and Blood Banks: Coordinate blood requests and availability.

NGOs and Financial Contributors: Provide verified medical and financial assistance.

Ambulance Services: Support emergency coordination and hospital access.

Community and Campus Volunteers: Promote blood donation and spread awareness.

3. Who Benefits if the Problem Is Solved?

Patients: Gain faster access to potential donors and blood-bank resources.

Families: Reduce stress from repeated calls and scattered messages.

Hospitals: Improve coordination and reduce manual effort.

Donors: Receive relevant, verified notifications.

Society: Benefits from stronger community participation and better emergency preparedness.

TEAM
HACKNOVA

CURRENT SITUATION & CONTEXT

GENESIS

Existing scenario, background and how things work today

Existing Scenario and Background

- In an emergency, families arrange blood and money mostly through personal contacts and social media
- Hospitals, blood banks, donors, and fundraising channels work independently of one anotherThe family handles logistics at the very moment they are least able to

Current Workflow

- The patient is admitted, and the doctor advises on blood and treatment costs
- The family phones relatives, friends, and colleagues, The request is forwarded across WhatsApp groups and social media
- Hospitals and blood banks are contacted one by one Replies arrive late, repeat each other, or cannot be verifiedFunds are raised separately through loans, appeals, or crowdfunding
- Hours are lost, and treatment may be delayed

Relevant Data or Facts

- Blood cannot be stored for long, and rare groups and platelets are especially hard to find at short notice
- Many donors are willing to help but never learn that someone nearby needs them
- Online medical fundraising exists, but it takes time to earn trust and is not linked to the patient's immediate need(Insert one or two verified figures here, with the source)

KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

Main difficulties and bottlenecks

1. The Emergency Time Gap - Delays in identifying available, potentially eligible blood donors during time-critical medical situations.

2. Fragmented Donor Information - Incomplete, outdated or scattered records of blood groups, donation history, hemoglobin levels and donor availability.

3. The Dual Emergency Burden - Families must coordinate blood requirements and urgent financial assistance through separate networks and platforms.

4. Rare Blood Group Accessibility - Difficulty locating donors with rare blood groups or specific blood characteristics within a clinically useful distance during emergencies.

5. The Donor Availability Uncertainty - Registered donors may be unreachable, unavailable, geographically distant or temporarily ineligible to donate.

KEY CHALLENGES

Root Causes Behind Them

1. Disconnected Healthcare Networks

- Donor communities, hospitals, blood banks and charitable organizations often operate through separate systems without a unified coordination mechanism.

2. Absence of Dynamic Health Records

- Donor profiles may lack updated donation dates, health measurements, availability status and verified location information.

3. Manual Emergency Coordination

- Reliance on phone calls, social media messages and personal contacts creates repetitive communication and delays in reaching suitable donors or contributors.

4. Limited Standardized Verification

Inconsistent document checks, identity validation and medical-request verification make it difficult to establish reliability across informal networks.

5. Lack of Intelligent Matching and Tracking

- Many informal systems do not combine blood-group requirements, geographical proximity, donor availability, donation intervals and request-status updates in one workflow.

KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

Consequences of Leaving Them Unsolved

1. Increased Risk During Medical Emergencies - Delayed access to appropriate blood supplies can complicate emergency care and may worsen patient outcomes.

2. Delayed Treatment and Financial Hardship - Difficulties securing blood and funds may disrupt treatment arrangements and increase pressure on already distressed families.

3. Underutilization of the Donor Community - Willing and potentially eligible donors may remain undiscovered while families repeatedly contact the same limited network of people.

4. Erosion of Public TrustFraudulent appeals, misleading donor information or poor financial transparency can discourage genuine donors and contributors from participating.

5. Unequal Access to Emergency SupportPatients with fewer personal contacts, limited digital access or weaker financial resources may face greater difficulty obtaining timely assistance.

EXISTING SOLUTIONS / APPROACHES

What is already being done, and where it falls short

Existing Methods, Systems or Alternatives

- 1. e-RaktKosh: Government platform for blood-bank information and blood availability.
- 2. Milaap & Ketto: Crowdfunding platforms that help patients raise funds for medical emergencies.
- 3. WhatsApp & Social Media: Informal networks for circulating urgent blood requests and fundraising appeals.
- 4. Hospital Blood Banks: Healthcare facilities that coordinate blood availability, donor screening and transfusion services.
- 5. Standalone Blood Donor Apps: Digital platforms that help users register donors and search by blood group or location.

EXISTING SOLUTIONS / APPROACHES

What is already being done, and where it falls short

Limitations of Each

Government blood portals: Blood-stock information does not guarantee immediate access to an available donor.

2. Crowdfunding platforms: Financial fundraising is separate from real-time blood donor coordination.

3. Social media appeals: Requests can be duplicated, outdated or difficult to verify.

4. Hospital blood banks: Families may need to contact multiple facilities to identify available supplies.

5. Standalone donor databases: Donor availability and current medical eligibility may not be updated.

The Gap They Leave Open — Key Challenges

1.Fragmented emergency support: Blood assistance and financial aid are coordinated through separate systems.

2. Uncertain donor readiness: Families lack a consolidated view of donor availability and donation history.

3. Delayed coordination: Manual communication can slow the process of connecting patients with potential donors.

4. Verification and transparency gaps: Families need reliable verification of emergency requests and clear fundraising updates.

5. No unified emergency dashboard: Blood donor responses, hospital coordination and crowdfunding progress are not consolidated into one workflow.

PROBLEM INSIGHTS & OPPORTUNITY AREAS

Resources required to solve the problem

- Hardware Requirements- Smartphones for donors, families, hospitals- Cloud servers for real-time alerts
- Software Requirements- Android/iOS app (Flutter) with Firebase backend- Google Maps API for donor distance and ETA- SMS/push alerts and UPI payment gateway
- Other Resources (Including Human Resources)- Developers, UI/UX designer, medical advisor- Hospital and blood bank tie-ups for verification- IMA, Rotary, Lions, NGOs for fundraising support

INITIAL DIRECTION & EXPECTED OUTCOME

What Is Your Expected Outcome?

1. What Our Team Intends to Do

Build a centralised platform connecting patients, blood donors, hospitals, and blood banks across India.

Enable verified blood requests with details such as blood group, hospital location, urgency, and donor availability.

Integrate verified medical crowdfunding to help families manage emergency expenses.

2. What We Aim to Improve

Reduce dependence on scattered phone calls, social media messages, and manual donor searches.

Improve coordination between patients, families, donors, hospitals, and blood banks.

Increase trust through verified blood requests, updated availability, and transparent fundraising.

3. What Successful Improvement Should Achieve

Help families identify potential donors and available blood-bank resources faster.

Reduce the stress and effort involved in arranging blood donations and financial assistance.

Create a scalable platform with scope for future AI-assisted donor prioritisation and blood-demand forecasting.

TEAM CONTRIBUTION

Who will handle which part of the execution

Team Member	Role	Responsibility in Execution
Pranayaa AS	Role	What they will handle
Pratibha B	Role	What they will handle
Visurtha N	Role	What they will handle
MK Kesava Pritha	Role	What they will handle